and the time-reversal symmetry

$$\mathscr{F}_t \circ \boldsymbol{R} \circ \mathscr{F}_t = \boldsymbol{R}$$

have a formal similarity, but they are essentially unrelated. A method can be symplectic but not time-reversible (e.g. Symplectic Euler) or it can be time-reversible and not symplectic (e.g. Trapezoidal Rule). However there are intriguing relationships between these properties which we discuss here.

Interestingly the time-reversal symmetry provides analogous properties in some respects to the properties of Hamiltonian systems. For example, consider a linear Hamiltonian system $\mathrm{d}\boldsymbol{z}/\mathrm{d}t = \boldsymbol{JAz}$, with $\boldsymbol{A}$ a symmetric matrix. If $\lambda$ is an eigenvalue of $\boldsymbol{JA}$ then $\boldsymbol{JAu} = \lambda\boldsymbol{u}$ for some eigenvector $\boldsymbol{u} \neq 0$. Because the matrix $\boldsymbol{JA}$ is real, we know that $\bar{\lambda}$ will also be an eigenvalue. At the same time, we know that since $\lambda$ is an eigenvalue of $\boldsymbol{JA}$ it is also an eigenvalue of its transpose $(\boldsymbol{JA})^T = \boldsymbol{A}^T\boldsymbol{J}^T = -\boldsymbol{AJ}$, thus

$$-\boldsymbol{AJu} = \lambda\boldsymbol{u}$$

multiplying by $\boldsymbol{J}$ and setting $\boldsymbol{v} = \boldsymbol{Ju}$ we have

$$-\boldsymbol{JAv} = \lambda\boldsymbol{v}$$

implying that $-\lambda$ (and hence also $-\bar{\lambda}$) is an eigenvalue of $\boldsymbol{JA}$. Real eigenvalues of $\boldsymbol{JA}$ are paired with their negatives. If the imaginary part is nonzero, the eigenvalues occur in quadruplets $\{\pm\lambda, \pm\bar{\lambda}\}$. The flow map is

$$\mathscr{F}_t(\boldsymbol{z}) = e^{t\boldsymbol{JA}}\boldsymbol{z}$$

and the exponential matrix will inherit a related structure within the spectrum: $\lambda$ an eigenvalue of $\exp(t\boldsymbol{JA})$ implies that $\bar{\lambda}$, $1/\lambda$ and $1/\bar{\lambda}$ are all eigenvalues of $\exp(t\boldsymbol{JA})$. This eigenvalue structure is generic for linear symplectic maps in general.

Suppose now we have a linear time-reversible map $\Phi(\boldsymbol{z}) = \boldsymbol{Tz}$, then

$$\boldsymbol{T}^{-1} = \boldsymbol{RTR}.$$

Given an eigenvalue, eigenvector pair $(\lambda, \boldsymbol{u})$ of $\boldsymbol{T}$, let $\boldsymbol{u} = \boldsymbol{Rv}$, so that $\boldsymbol{Ru} = \boldsymbol{R}^2\boldsymbol{v} = \boldsymbol{v}$, then

$$\boldsymbol{T}^{-1}\boldsymbol{v} = \boldsymbol{RTRv} = \boldsymbol{RTu} = \lambda\boldsymbol{Ru} = \lambda\boldsymbol{v}.$$

Thus $\lambda$ is an eigenvalue of $\boldsymbol{T}^{-1}$ which, in turn, implies that $1/\lambda$ is an eigenvalue of $\boldsymbol{T}$. The matrix being real implies that the conjugates of $\lambda$ and $1/\lambda$ are also eigenvalues, thus we have the same eigenvalue quadruplets as for a linear symplectic map.

Recall that a pair of maps $\boldsymbol{\Phi}$ and $\boldsymbol{\Psi}$ are said to be conjugate if there is a homomorphism $\boldsymbol{\chi}$ such that